Valentina **Astore**

PhD Student at INRIA Saclay & École polytechnique

Bâtiment Alan Turing - Campus de l'École polytechnique
1 rue Honoré d'Estienne d'Orves, 91120 Palaiseau, France

valentina.astore@inria.fr · valentina-astore.github.io · linkedin.com/in/valentina-astore ·
Age: 26 · Nationality: Italian

Education

PhD in Computer Science — *Palaiseau, France*
INRIA Paris-Saclay & École Polytechnique — *Oct. 2024 - Present*

- Research areas: Algebraic Coding Theory, Finite Fields, Finite Geometry, and applications to Cryptography.
- Supervisors: Alain Couvreur and Martino Borello.

MSc in Mathematics (Cryptography & Coding Theory) — *Trento, Italy*
University of Trento — *Sept. 2021 - March 2024*

- Highly selective master program (10 students per year, with admission test).
- Graduation grade: 110 with honour/110.

Master's Project — *Saint-Denis, France*
University of Paris 8 - LAGA — *Oct. 2023 - Dec. 2023*

- Topics: Coding Theory, Rank-Metric Codes, Distinguishers, Castelnuovo-Mumford Regularity.
- Supervisors: Martino Borello and Marco Calderini.

Publications

Preprints

V. Astore, F. Zullo. **On the Length of Maximum Sum-Rank Distance Codes.** *arXiv:2610.23633*

V. Astore, M. Borello, A. Couvreur, F. Salizzoni. **A q-analogue of the rational normal curve and linearized Reed-Solomon codes.** *arXiv:2606.13246*

Journal Papers

V. Astore, M. Borello, M. Calderini, F. Salizzoni. **A geometric invariant of linear rank-metric codes.** *SIAM Journal on Applied Algebra and Geometry.* 2025

Complementary Activities

Talks

Invited talk. 07-11 September 2026, Messina (Itay).
2nd Joint Meeting Brazil-Italy in Mathematics.

Contributed talk. 25-29 May 2026, Naples (Italy).
Combinatorics 2026.

Contributed talk. 23-26 February 2026, Bordeaux (France).
OpeRa 2026 - Open Problems on Rank-Metric Codes.

Contributed talk. 15 September - 19 September 2025, Brussels (Belgium) .
Finite Geometry and Friends - A Brussels summer school on Finite Geometry.

Invited talk. 07-11 July 2025, Madison (Wisconsin, US).
SIAM Conference on Applied Algebraic Geometry (AG25).

Contributed talk. 31 March - 04 April 2025, Pornichet (France).
Journées Codage et Cryptographie (C2).

Contributed talk. 14-16 February 2024, Caserta (Italy).
OpeRa 2024 - Open Problems on Rank-Metric Codes.

Workshops

COSMO 2025. 03-05 February 2025, Paris (France).
Topics: Combinatorics of spaces under multiplication and applications.
Member of the organizing committee.

PQCifris 2022 School on Post-Quantum Cryptography. October 2022, Trento (Italy).
Topics: Introduction to post-quantum cryptography, multivariate-based, code-based, isogeny-based, and lattice-based methods.

De Cifris Trends in Modern Cryptography. March 2022 - May 2022, Trento (Italy).
Topics: Post-quantum cryptography, cloud encryption and secure multiparty computation.

Academic Teaching

Teaching Assistant

Introduction to Computer Programming. A.Y. 2025/26 and 2026/27.
CSC_1F001_EP course at École polytechnique

Cryptography. A.Y. 2025/26.
CSC_5CY08_TA course at ENSTA Paris

Foundations of Computer Science: Logic, Models, Computability and Computational Complexity. A.Y. 2024/25.
CSC_41012_EP - INF 412 course at École polytechnique

Professional Experience

Substitute Teacher *Bassano del Grappa (VI), Italy*
Scuola Primaria Marconi *May 2024*
- On-call work
- Sector: Education (primary school).

Research and Development Programmer *Milano, Italy*
BV TECH S.p.A. *March 2023 - July 2023*
- Internship experience, held remotely.
- Sector: IT Services and IT Consulting.
- Tasks: Analyze and improve the security strength of a communication service for messages and VoIP calls.

Administrative assistant *Torquay, United Kingdom*
Essential 6 Ltd *August 2017- Sept. 2017*
- Internship experience.
- Sector: First Aid & Safety Training .
- Tasks: Management of electronic and paper mail, production and revisions of costumer's certificates, organization of team building events for the staff.